

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

(197511415)

I Kanish Kumar S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S. Kanish Kumar

(94)

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CO4 - AT2

ARTICLE REVIEW

SELECTED RESEARCH ARTICLE :

AUTHORS : Jonathan Bodine and Dorit S. Hochbaum.

TITLE : A Better Decision Tree: The Max-Cut Decision Tree with Modified PCA Improves Accuracy and Running Time.

JOURNAL : SIAM Computer Science, Volume 3, Article 313.

YEAR : 2022

DOI : 10.1007/s42979-022-01147-4

DOMAIN: Machine learning / Classification.

ML PROBLEM: Improving Decision Tree accuracy and efficiency.

The article proposes a Max-Cut Splitting Criterion combined with Node means PCA to improve Decision Tree, especially for high-dimensional datasets.

TASK 1 : ARTICLE SUMMARY

(a) DOMAIN AND SPECIFIC ML PROBLEM :

The research focuses on Machine Learning and Data Classification. It addresses limitations of traditional CART Decision Trees, particularly their accuracy and computational efficiency on high-dimensional data.

CART uses ~~Gini~~ impurity for splitting, but this may not fully consider distances between observations. The authors propose Max-Cut splitting and Node Means PCA to create more meaningful local feature representation.

B) ARTICLE OBJECTIVE AND RESEARCH GAP:

The objective is to improve Domain Tree accuracy and Speed by modifying the Splitting Criterion and feature representation. The research gap is that Gini impurity mainly focuses on class separation and does not fully use feature - space distances.

The proposed Max Cut Node Means PCA method applies PCA locally at each tree node. Experiments on over 20000 Synthetic datasets and several real datasets showed improved accuracy and reduced runtime, especially for high-dimensional problems.

TASK 2 : METHODOLOGY ANALYSIS

a) TECHNIQUES AND DATASET:

Techniques used,

- * CART Decision Tree
- * Gini Impurity
- * Max-cut
- * PCA
- * Node Means PCA

The study used 20,000+ Synthetic datasets and real datasets including Iris, Wine Quality, ILPD, MNIST, CIFAR-10 and CIFAR-100.

Synthetic datasets:

- * Over 20000 datasets.
- * Training sizes from 100 to 300000.
- * Binary and 10 class classification.
- * Standardized using standard scalar.

smaller Datasets used 10x10 cross validation, while larger datasets used an 80:20 train test split.

B) MAPPING TO CO4 TOPICS:

CO4 TOPIC	CONNECTION
Decision Trees	Improves CART Decision Trees.
Inductive learning	Learns rules from labelled data.
Statistical learning	used accuracy, confidence intervals and significance tests.
ML Evaluation	compares accuracy, runtime and statistical results.

TASK 3: RESULTS & CRITICAL EVALUATION

a) KEY RESULT:

MODEL / METHOD	MAIN RESULT
CART Gini	Lower performance on many high dimensional datasets.
Gini + Node Means PCA	Improved accuracy
Max cut + Node Means PCA	strong accuracy and runtime
Maxcut Tree	Faster on large datasets
CIFAR-100	49.4% accuracy improvement
CIFAR-100 Runtime	About 6.8 x Faster

9) CRITICAL EVALUATION:

The results are convincing because of the large number of experiments, real-world datasets and statistical testing.

However, synthetic datasets may not reflect all real-world conditions. Different dataset sizes and evaluation methods also make direct comparisons difficult.

9) REAL - WORLD APPLICATIONS:

Possible applications include,

- * Image classification
- * Medical diagnosis
- * Fraud detection
- * Cybersecurity
- * Scientific data analysis.

The method maintains Decision Tree interpretability while improving performance. However, it may require substantial computational resources and may be less effective with very noisy data.

TASK 4 : REFLECTION & LEARNING OUTCOME :

9) THREE KEY INSIGHTS:

1. Splitting criteria matter,

The splitting criterion strongly affects Decision Tree performance, Max-cut uses both class differences and observation distances.

CONCLUSION : Decision Tree Learning

Local PCA can improve classification.

Node Means PCA transforms features locally at each tree node, helping the model identify useful directions for each subproblem.

CO4 Connection: Statistical learning and dimensionality reduction

3. Accuracy and efficient are both important,

A strong ML model should provide high accuracy while using reasonable computational resources.

CO4 Connection: Model evaluation and performance analysis.

B) PROPOSED EXTENSION :

A useful extension would test Max Cut Node Means PCA on Cyber Security network-traffic datasets.

The comparison could include,

- * CART
- * Random Forest
- * Max Cut Decision Tree
- * Max Cut Node Means PCA

This experiment would test whether the method is effective for -high-dimensional intrusion and threat-classification data.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

$$20 + 20 + 25 + 15 + 20$$

$$= 100$$

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